

06 — API Reference

Base URL (local): `http://localhost:8000`
API prefix: `/api`
Interactive docs: `http://localhost:8000/docs`

Health & static

Method	Path	Notes
GET	<code>/health</code>	Liveness + backend meta
GET	<code>/storage/...</code>	Static serving of uploads, masks, outputs, textures

Ingestion

Method	Path	Body	Response
POST	<code>/api/ingestion/upload</code>	multipart/form-data field <code>file</code>	<code>project_id</code> , image URL, size, warnings
GET	<code>/api/ingestion/projects/{project_id}</code>		Project metadata

Rejects: unreadable image, wrong type, oversize, below minimum resolution.
Warns (still accepts): blurry, too dark, too bright.

Segmentation

Method	Path	Body	Response
POST	<code>/api/segmentation/{project_id}</code>	<code>{ "categories": string[], "model"?: "grounded_sam" }</code>	RegionMap list (masks, polygons, areas)
GET	<code>/api/segmentation/{project_id}/regions</code>		Current RegionMap
PUT	<code>/api/segmentation/{project_id}/regions/{category_id}</code>	Masked image payload (edited mask)	Updated region

Empty `categories` → defaults: wall, balcony, rooftop, gate, window, door.

Materials & rates

Method	Path	Body	Response
GET	/api/materials	—	Catalog (paint + textures, rates, units)
PUT	/api/materials/rates	Rate update list	Confirmation

Meta

Method	Path	Response
GET	/api/meta/categories	Default + full category taxonomy
GET	/api/meta/models	Available segmentation backends
POST	/api/meta/models/activate	Activate model key
GET	/api/meta/render-modes	Available render modes (classical)
GET	/api/meta/model-status	Download / readiness of weights

Rendering

Method	Path	Body	Response
POST	/api/rendering/{project_id}	{ "selections": [{ category, material_key, color? }], "mode"?: "classical" }	input_url, output_url, applied list

Requires segmented regions for selected categories.

Costing

Method	Path	Body	Response
POST	/api/costing/{project_id}/estimate	{ selections, facade_width_m, facade_height_m, optional waste }	Lines, subtotal, total INR, disclaimer

See [04_Estimation_and_Costing.md](#) for formulas.

Reporting

Method	Path	Body	Response
POST	/api/reporting/{project_id}/pdf	selection + facade dims (same spirit as costing)	PDF download

Requires a prior successful render (`output_url` present).

Typical error codes

Code	Meaning
404	Unknown project
422	Unusable upload, missing region, unknown material, empty selections, no render for PDF

RegionMap fields (serialization)

Field	Description
category	Facade label
mask_url	Tinted RGBA preview
polygons	Contours for overlays
pixel_area	Mask pixel count
instance_count	Merged detections
confidence	Mean detector score